STAT 527 Advanced Regression Analysis
Fall 2024

Problems with the Error

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What can go wrong with the errors?

So far, we have only considered the case where the Ys have equal variances and are uncorrelated. This follows from the assumption

$$\text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}$$

We also assume normal distribution for the tests. What about the following cases?

- Unequal variance
- Correlated
- Heavy-tailed

Generalized Least Squares

Consider the usual model equation

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where $\boldsymbol{X}$ is an $n \times p'$ matrix of rank p'.

We still assume that $\mathrm{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$, but now assume

$$\mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2\boldsymbol{\Sigma}$$

for some invertible matrix $\boldsymbol{\Sigma}$ (symmetric and positive definite) and some constant σ^2.

Unless $\boldsymbol{\Sigma}$ is a multiple of the identity matrix, the Gauss-Markov conditions are not satisfied, and minimizing the residual sum of squares does not necessarily result in best linear unbiased estimation.

Generalized Least Squares Estimation

In the method of **generalized least squares (GLS)**, we assume $\boldsymbol{\Sigma}$ is *known* (but σ^2 may still be unknown).

Instead of minimizing RSS, we minimize the **generalized residual sum of squares**,

$$GRSS(\boldsymbol{\beta}) = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$$

The **generalized least squares (GLS) estimator** is given by

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{y}$$

and it minimizes $GRSS(\boldsymbol{\beta})$ (Proof next slides).

Proof

To see this, note that $\boldsymbol{\Sigma}^{-1}$ can be written as follows (using a matrix decomposition):

$$\boldsymbol{C}^T\boldsymbol{C} = \boldsymbol{\Sigma}^{-1}$$

so that

$$\boldsymbol{C}^{-1}\boldsymbol{C}^{-T} = \boldsymbol{\Sigma}$$

(One such $\boldsymbol{C}$ satisfying these conditions is an $n \times n$ matrix called a **square root** of $\boldsymbol{\Sigma}^{-1}$.)

Note: $\boldsymbol{C}$ is a known constant matrix, because $\boldsymbol{\Sigma}$ is known and constant.

Proof Ctd - Transformed Model

Let's multiply the left and right sides of $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ by $\boldsymbol{C}$ ...

$$\boldsymbol{C}\boldsymbol{y} = \boldsymbol{C}\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{C}\boldsymbol{\varepsilon}$$

Define $\boldsymbol{z} = \boldsymbol{C}\boldsymbol{y}$, $\boldsymbol{M} = \boldsymbol{C}\boldsymbol{X}$, and $\boldsymbol{\delta} = \boldsymbol{C}\boldsymbol{\varepsilon}$.

Then we have the model

$$\boldsymbol{z} \;=\; \boldsymbol{M}\boldsymbol{\beta} + \boldsymbol{\delta}$$

where $\mathrm{E}(\boldsymbol{\delta}) = \mathbf{0}$ and

$$\begin{aligned} \mathrm{var}(\boldsymbol{\delta}) \;&=\; \boldsymbol{C}\,\mathrm{var}(\boldsymbol{\varepsilon})\boldsymbol{C}^T \;=\; \sigma^2\boldsymbol{C}\boldsymbol{\Sigma}\boldsymbol{C}^T \\ &=\; \sigma^2\boldsymbol{C}\boldsymbol{C}^{-1}\boldsymbol{C}^{-T}\boldsymbol{C}^T \;=\; \sigma^2\boldsymbol{I} \end{aligned}$$

Proof Ctd

Thus, this model satisfies the conditions, and is obtained from the least squares estimator

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (\boldsymbol{M}^T\boldsymbol{M})^{-1}\boldsymbol{M}^T\boldsymbol{z} \\ &= (\boldsymbol{X}^T\boldsymbol{C}^T\boldsymbol{C}\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{C}^T\boldsymbol{C}\boldsymbol{y} \\ &= (\boldsymbol{X}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{y}\end{aligned}$$

Moreover, this minimizes

$$\begin{aligned}(\boldsymbol{z}-\boldsymbol{M}\boldsymbol{\beta})^T(\boldsymbol{z}-\boldsymbol{M}\boldsymbol{\beta}) &= \big(\boldsymbol{C}(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta})\big)^T\big(\boldsymbol{C}(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta})\big) \\ &= (\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta})^T\boldsymbol{C}^T\boldsymbol{C}(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}) \\ &= GRSS(\boldsymbol{\beta})\end{aligned}$$

Generalized Least Squares Continued

Similarly, other GLS estimates:

$$\begin{aligned} \hat{\beta} &= (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y \\ var(\hat{\beta}) &= (X^T \Sigma^{-1} X)^{-1} \sigma^2 \end{aligned}$$

$$\hat{\sigma}^2 = \frac{\hat{\epsilon}^T \Sigma^{-1} \hat{\epsilon}}{n - (p' + 1)}$$

Weighted Least Squares

A special case of generalized least squares is **weighted least squares**. This arises when the errors are uncorrelated, but **heteroscedastic** (have different variances). For instance, we might have $\operatorname{var}(\varepsilon_i) = \sigma^2/w_i$ instead of homoscedastic errors, $\operatorname{var}(\varepsilon_i) = \sigma^2$.

In weighted least squares, $\boldsymbol{\Sigma}^{-1}$ has the form

$$\boldsymbol{\Sigma}^{-1} = \boldsymbol{W} = \begin{pmatrix} w_1 & & & 0 \\ & w_2 & & \\ & & \ddots & \\ 0 & & & w_n \end{pmatrix}$$

where w_i is called the **weight** of the ith observation.

Example Scenarios

Some circumstances in which weighted least squares may arise:

- y_i is actually an average of n_i equally variable observations (all with the same σ^2 for all i). Then $\mathrm{var}(\varepsilon_i) = \sigma^2/n_i$, and an appropriate weight is

$$w_i = n_i$$

- y_i is an average of $n_i > 1$ equally variable observations that have variance σ_i^2 (so $\mathrm{var}(\varepsilon_i) = \sigma_i^2/n_i$). Suppose their sample standard deviation s_i is available. Then $\mathrm{var}(\varepsilon_i) \approx s_i^2/n_i$, and an appropriate weight may be

$$w_i = n_i/s_i^2$$

Example: Travel Times

A data set was collected to investigate the relationship between anticipated travel times and actually reported travel times between selected zones in Chicago. Each case in the data set represents a pair of zones in Chicago.

The predictor X gives travel times computed from a formula: bus-timetables augmented by walk times from zone-centroids to bus-stops (assuming a walking speed of 3 m.p.h.) and expected waiting times for the bus.

The variable Y was the average travel time as reported to the U.S. Census Bureau by n travelers. However, n depends on the pair of zones being considered.

Is Y on average reflecting the anticipated time X?

```
> travel <- read.table("travel.dat", header=TRUE)

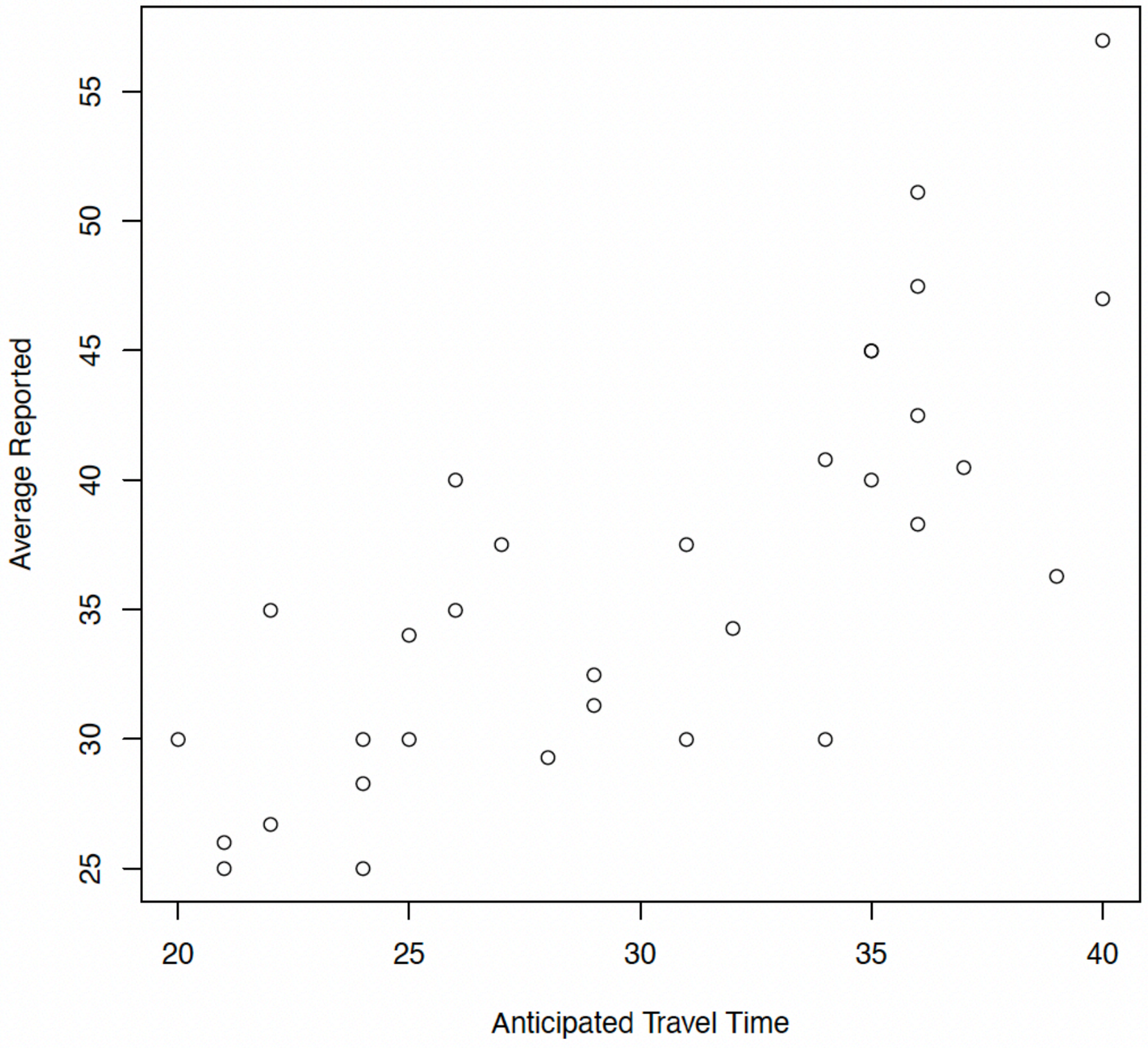
> # Examine data

> head(travel)
  n  x    y
1 1 26 35.0
2 1 40 57.0
3 7 32 34.3
4 3 36 38.3
5 2 27 37.5
6 4 39 36.3
```

```
> summary(travel)
       n                x               y
 Min.   : 1.000   Min.   :20.00   Min.   :25.00
 1st Qu.: 2.000   1st Qu.:24.75   1st Qu.:30.00
 Median : 3.000   Median :30.00   Median :35.00
 Mean   : 3.812   Mean   :30.00   Mean   :36.20
 3rd Qu.: 4.250   3rd Qu.:35.25   3rd Qu.:40.58
 Max.   :17.000   Max.   :40.00   Max.   :57.00

> # Plot data pairs

> plot(travel$x, travel$y, xlab="Anticipated Travel Time",
+      ylab="Average Reported")
```


Average Reported
25
30
35
40
45
50
55
20
25
30
35
40
Anticipated Travel Time

Unweighted vs Weighted Analyses

```
> # Fit unweighted least squares

> unweighted <- lm(y ~ x, data=travel)

> summary(unweighted)
...

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.0325     4.5137   1.336    0.191
x             1.0056     0.1475   6.819 1.45e-07 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

...

>   # t-test for slope=1

> t.stat <- (1.0056 - 1)/0.1475

> p.val <- 2*(1-pt(abs(t.stat), df.residual(unweighted)))

> p.val
[1] 0.9699662
```

why is number not 1.45e-07?

Unweighted vs Weighted Analyses

```
> # Fit weighted least squares (weight = number averaged)

> weighted <- lm(y ~ x, data=travel, weights=n)

> summary(weighted)
...

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.2932     4.5903   0.500    0.621
x             1.1319     0.1475   7.676 1.46e-08 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

...

>   # t-test for slope=1

> t.stat <- (1.1319 - 1)/0.1475

> p.val <- 2*(1-pt(abs(t.stat), df.residual(weighted)))

> p.val
[1] 0.3783134
```

Robust Regression

Main concern: Non-normal and heavy-tailed error distribution

- M-estimation
- Least trimmed squares

M-estimation

Find β to minimize

$$\sum_{i=1}^{n} L(y_i - x_i^T \beta)$$

$L(\cdot)$ is called the loss function.

M-estimation

Find β to minimize

$$\sum_{i=1}^{n} L(y_i - x_i^T \beta)$$

$L(\cdot)$ is called the loss function.

Possible loss functions:

- $L(z) = z^2$ least squares (LS)
- $L(z) = |z|$ least absolute deviations (LAD)
- Huber's method

$$L(z) = \begin{cases} z^2/2 & \text{if } |z| \leq c \\ c|z| - c^2/2 & \text{otherwise} \end{cases}$$

c is a robust estimate of σ, e.g., the median of $|\hat{\epsilon}_i|$.

Gala Example

Number of species of tortoise on the various Galapagos islands

- Response: number of species of tortoise
- Predictors: number of endemic species, area of the island, highest elevation of the island, distance from the nearest island, distance from Santa Cruz Island, area of the adjacent island

```
> data(gala)
## Least squares
> g <- lm(Species ~ Area + Elevation + Nearest
    + Scruz + Adjacent, data=gala)
> summary(g)
Coefficients:
             Estimate Std.Error t value Pr(>|t|)
(Intercept) 7.068221 19.154198   0.369 0.715351
Area       -0.023938  0.022422  -1.068 0.296318
Elevation   0.319465  0.053663   5.953 3.82e-06
Nearest     0.009144  1.054136   0.009 0.993151
Scruz      -0.240524  0.215402  -1.117 0.275208
Adjacent   -0.074805  0.017700  -4.226 0.000297

Residual standard error:   60.98 on 24 degrees of free
Multiple R-Squared: 0.7658     Adjusted R-squared: 0.7
F-statistic:  15.7 on 5 and 24 DF      p-value: 6.838e
```

```
## Huber's method
> library(MASS)
> ghuber <- rlm(Species ~ Area + Elevation + Nearest
    + Scruz + Adjacent, data=gala)
> summary(ghuber)
Coefficients:
             Value   Std.Error t value
(Intercept)  6.3611 12.3897    0.5134
Area        -0.0061  0.0145   -0.4214
Elevation    0.2476  0.0347    7.1320
Nearest      0.3592  0.6819    0.5267
Scruz       -0.1952  0.1393   -1.4013
Adjacent    -0.0546  0.0114   -4.7648
Residual standard error:  29.73 on 24 degrees of freed
```

Summary: Robust methods

- Protect against outliers and heavy tails... but not misspecified structure (model or error)
- If robust and LS fits are very different, cause to worry
- Useful when automatic fitting is needed (no human intervention)

Thank you!

We gratefully acknowledge the previous course instructor Naveen Narisetty and various online sources for the images and slides presented.